# Aanchal Vij

Assistant Professor | Artificial Intelligence & Medical Imaging Researcher

Department of Computer Science and Engineering, IILM University

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## Research Profile

Assistant Professor in Computer Science and Engineering with over ten years of teaching and academic experience. Researcher specializing in Artificial Intelligence, Machine Learning, Deep Learning, Medical Image Processing, Computer Vision, and Healthcare AI, with a primary research focus on lung cancer detection and classification using CT images. Research experience includes hybrid deep-learning architectures, attention mechanisms, Transformer-based models, image preprocessing, texture analysis, optimization, and data-driven healthcare applications.

**Research direction:** Explainable and trustworthy AI for medical imaging, with emphasis on clinically meaningful, robust, and generalizable lung cancer detection systems.

## Research Interests

- Artificial Intelligence and Machine Learning
- Deep Learning and Computer Vision
- Medical Image Processing and CT Image Analysis
- Lung Cancer Detection and Classification
- Healthcare Artificial Intelligence
- Explainable and Trustworthy AI
- Attention Mechanisms and Transformer Architectures
- Image Enhancement, Feature Extraction, and Optimization

## Academic Employment

**Assistant Professor, Department of Computer Science and Engineering** — IILM University
Greater Noida, India — *January 2026 – Present*

- Teaching undergraduate Computer Science and Engineering courses.
- Teaching and academic responsibilities across Data Mining and Warehousing, Theory of Computation and related computing subjects.
- Course coordination, lecture planning, practical/laboratory development, question-bank preparation, and student mentoring.
- Research supervision and mentoring in Artificial Intelligence, Machine Learning, and medical image analysis.

**Previous Academic Experience**

**Assistant Professor** — Sharda University
Greater Noida, India — *September 2024 – January 2026*

**Assistant Professor** — G.L. Bajaj Institute of Technology and Management
Greater Noida, India — *October 2023 – September 2024*

**Assistant Professor** — Galgotias University
Greater Noida, India — *August 2019 – October 2023*

**Assistant Professor** — Harlal Institute of Management and Technology
Greater Noida, India — *August 2016 – February 2017*

**Assistant Professor** — S.S.D. Women Institute of Technology
Bathinda, Punjab — *August 2015 – May 2016*

## Education

**Ph.D., Computer Science and Engineering** — Galgotias University
*Thesis submitted in September 2026; degree award pending.*
**Thesis:** A Novel Methodology of Early Lung Cancer Prediction Using Soft Computing

- Research focused on lung cancer prediction/detection using medical imaging, soft computing, machine learning, and deep learning.
- Research included CT-image preprocessing, texture analysis, optimization, hybrid machine-learning models, and deep-learning architectures.

**M.Tech., Computer Science and Engineering** — RC-GNDU, Jalandhar

2015 CGPA: 8.50

**B.Tech., Computer Science and Engineering** Doaba Group of Institutions, Kharar

2013 Percentage: 77%

**Diploma, Computer Science** S.R.S. Government Polytechnic for Women, Ludhiana

2010 Percentage: 69.8%

## Research Contributions

- Development and evaluation of hybrid AI architectures for lung cancer classification from CT images.
- Investigation of EfficientNetV2-S, CBAM attention, and Transformer encoders for local feature extraction, attention-driven refinement, and global dependency modelling.
- Application of CLAHE, wavelet denoising, normalization, augmentation, SMOTE, and focal-loss strategies for medical-image learning.
- Investigation of GLCM texture features and optimization-assisted feature selection within soft-computing and hybrid machine-learning frameworks.
- Evaluation using accuracy, precision, recall, F1-score, AUC, and statistical comparison methods.

## Selected Research Projects

**Hybrid Deep Learning Framework for Lung Cancer Detection**
EfficientNetV2-S + CBAM + Transformer-based architecture for multi-class classification of CT images using the IQ-OTHNCCD dataset.

**Soft Computing Framework for Lung Cancer Classification**
GLCM-based texture representation combined with optimization and hybrid stacked machine-learning models for multi-class lung cancer classification.

**Explainable and Trustworthy Healthcare AI – Future Research Direction**
Development of interpretable, robust, and generalizable AI methods for medical image analysis, with emphasis on lung cancer screening and decision support.

## Publications

| No. | Year | Title | Venue / Publication Type | Authors | DOI / Identifier |
|---|---|---|---|---|---|
| 1 | 2026 | Hybrid CNN-Based Transformer Pipeline With Radiomic Fusion for Multi-Class Lung Cancer Detection | Journal of Computer Science, 22(6), 1785–1796 | Aanchal Vij, Kuldeep Singh Kaswan, Anand Nayyar | 10.3844/jcssp.2026.1785.1796 |
| 2 | 2026 | Machine Learning Advancements for Lung Cancer Detection: An In-Depth Review and Future Prospects | Journal of Computer Science, 22(1), 260–272 | Aanchal Vij, Kuldeep Singh Kaswan, Anand Nayyar | 10.3844/jcssp.2026.260.272 |
| 3 | 2026 | From Detection to Decision Support: A Hybrid CNN-Transformer Model for Early Lung Cancer Screening | IEEE IC2PCT 2026, conference paper | Aanchal Vij, Kuldeep Singh Kaswan, Anand Nayyar | 10.1109/IC2PCT68894.2026.11584483 |
| 4 | 2026 | AI-Powered Resume and Job Description Matcher: A Semantic Similarity Approach Using SBERT and Contrastive Fine-Tuning | IEEE ICSFT 2026, conference paper | Jitendra Singh, Mekhala, Kamini Tanwar, Ashish Kumar, Aanchal Vij, Ashish Jain | 10.1109/ICSFT66733.2026.11507545 |
| 5 | 2026 | Bridging Fiber Optic and Free-Space Optical Communication: A Comparative Study and Hybrid Framework using Artificial Intelligence | IEEE I3CTCON 2026, conference paper | Aanchal Vij, Priyanka Gaba, Sourav Vij | 10.1109/I3CTCON68242.2026.11507287 |
| 6 | 2026 | Insect and Pest Recognition: A Lightweight Adaptive Feature-Level Ensemble Learning Approach | IEEE conference paper | Aanchal Vij and collaborators | 10.1109/GTSS68743.2026.11679638 |
| 7 | 2026 | An Interpretable CNN Ensemble Framework for Multi-Class Lung Cancer Classification | IEEE conference paper | Aanchal Vij and collaborators | 10.1109/GTSS68743.2026.11679541 |

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| 8 | 2026 | Benchmarking Pretrained Language Models as Pseudo-Label Annotators for Financial Sentiment Analysis | IEEE conference paper | Aanchal Vij and collaborators | 10.1109/GTSS68743.2026.11679744 |
| 9 | 2026 | DSMNet-V3: Lightweight Dual-Stream ROI-Focused MobileNetV3 Network for Driver Drowsiness Detection | IEEE conference paper | Aanchal Vij and collaborators | 10.1109/GTSS68743.2026.11679671 |
| 10 | 2025 | An Enhanced CT-Based Lung Cancer Classification Framework Using a Hybrid EfficientNetV2, CBAM, and Transformer Architecture | IEEE ACET 2025, conference paper | Aanchal Vij, Kuldeep Singh Kaswan, Anand Nayyar | 10.1109/ACET67282.2025.11430402 |
| 11 | 2025 | Multilevel Optimization with Hybrid Stack Model for Lung Cancer Classification | Journal of Information Systems Engineering and Management, 10(28s), 993–1019 | Aanchal Vij, Kuldeep Singh Kaswan, Anand Nayyar | 10.52783/jisem.v10i28s.4762 |
| 12 | 2025 | Machine Learning-Based Image Analysis for Cotton Crop Health: A Comparative Study | IEEE MRIE 2025, conference paper | Ankit Yadav, Vinay Sharma, Aanchal Vij, Jitendra Singh, Amit Chaudhary | 10.1109/MRIE66930.2025.11156686 |
| 13 | 2025 | Multi-Modal Plagiarism Detection in the Era of AI: A Comprehensive Review and Novel Framework Proposal | IEEE ACET 2025, conference paper | Anshu Kumar, Purushottam Choudhary, Anirban Dutta, Aanchal Vij | 10.1109/ACET67282.2025.11430248 |
| 14 | 2024 | Comparative Analysis of Web APIs: RESTful and GraphQL | Lecture Notes in Electrical Engineering, 1232, 159–169 | Aparna Seth, Vinay Kumar, Vaibhav Kulshrestha, Aanchal Vij | 10.1007/978-981-97-5231-7_14 |
| 15 | 2023 | Prediction of Lung Cancer using Convolution Neural Networks | IEEE AISC 2023, pp. 737–741 | Aanchal Vij, Kuldeep Singh Kaswan | 10.1109/AISC56616.2023.10085058 |
| 16 | 2023 | Artificial Intelligence and Regulatory Framework for Sustainable Plastic Management: An Analytical Study towards Achieving SDGs | IEEE ICCSAI 2023 | Isha Pandey, Sandhya Kumari, Aanchal Vij | 10.1109/ICCSAI59793.2023.10421531 |
| 17 | 2022 | Lung cancer detection using scan images | IEEE ICAC3N 2022 | Kartik Kumar, Shivank Srivastava, Aanchal Vij | 10.1109/ICAC3N56670.2022.10074331 |
| 18 | 2022 | Comparative analysis of Workflow Management System using Agents and Multi-agents in Colored Petri Nets | IEEE ICDCECE 2022 | Aanchal Vij, Kuldeep Singh Kaswan, Chinar Bawa | 10.1109/ICDCECE53908.2022.9792729 |
| 19 | 2022 | Case Study on Online Fraud Detection using Machine Learning | IEEE ICACITE 2022 | Ayushi Pandey, Harsh Jaiswal, Aanchal Vij, Tushar Mehrotra | 10.1109/ICACITE53722.2022.9823538 |
| 20 | 2022 | An Instructional Approach to Power System Operations Utilizing Load Flow Simulations | IEEE INCOFT 2022 | Upasana Rani, Aanchal Vij | 10.1109/INCOFT55651.2022.10094416 |
| 21 | 2021 | Research Problems in Recommender Systems | Journal of Physics: Conference Series, 1717, 012002 | Nitin Mishra, Saumya Chaturvedi, Aanchal Vij, Sunita Tripathi | 10.1088/1742-6596/1717/1/012002 |
| 22 | 2014 | A Review of Handoff and Location Management in Ad-Hoc Networks | ICCCS 2014 conference publication | Aanchal Vij and Jyooteesh Malhotra | |
| 23 | 2015 | A Review on Agent Oriented Software Engineering | International Journal of Advanced Research in Computer and Communication Engineering | Aanchal Vij and collaborators | 10.17148/IJARCCE.2015.4495 |

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| 24 | 2015 | Metrics for Project Estimation in Software Engineering: A Review | International Multi Track Conference (IMTC) | Aanchal Vij and collaborators | pp. 187–190. |
| 25 | 2015 | A Review on Genetic Algorithm to Deal with Optimization of Parameters of Constructive Cost Model | International Journal of Advanced Research in Computer and Communication Engineering | Aanchal Vij and collaborators | 10.17148/IJARCCE.2015.4491 |
| 26 | 2015 | A Study on Tools Used to Implement Agent Oriented Software Engineering | International Journal of Innovative Research in Computer and Communication Engineering | Aanchal Vij and collaborators | pp. 150-155 |

## Patents

- **Systems and Methods for IoT Enabled Detection of Leakage in Gas Distribution Pipe Network.**
- **Automated IoT Based Smart Agriculture Monitoring and Protection System with Drone.**
- **A System Method for Physiotherapy Treatment for Osteoarthritis based on IoT, AI, ML.**

## Book Chapters

- **Machine Learning and Deep Learning Algorithms for IoD.** In *The Internet of Drones*, CRC Press.
- **Assessing Awareness and Accessibility of Government Welfare Schemes Among Citizens: A Policy Implementation Perspective.** In *Computing, Communication, and Sustainable Energy Technologies*, Chapter 47, CRC Press. DOI: 10.1201/9781042018222-47.
- **An Integrative Review of Deep Learning-Based Approaches for Indian Sign Language to Speech Translation.** N. Singh, S. Singh, and A. Vij. In *Computing, Communication, and Sustainable Energy Technologies*, pp. 269–274, CRC Press.

## Academic Service and Reviewing

- Reviewer for academic journals and conferences in Computer Science, Artificial Intelligence, Machine Learning, and related areas.
- Academic and conference review activities.
- Member of institutional committees including NAAC Criteria 5 Committee and Placement Committee in previous academic roles.

## Professional Memberships

- **IEEE** – Member
- **IEEE Women in Engineering (WIE)** – Member
- **Association for Computing Machinery (ACM)** – Member
- **Computer Society Teachers Association (CSTA)** – Member
- **International Association of Engineers (IAENG)** – Member
- **IFERP** – Member
- **The Institute of Engineers** – Member

## Selected Faculty Development and Research Training

- FDP on High-Performance Computing and AI Workshop, June 2021.
- E&ICT FDP on Machine Learning and Deep Learning with Python (MLDLP), June 2021.
- FDP on R, February 2021.
- Faculty Development Programme on Moodle Learning Management System, December 2020.
- Indo-Muscat One Week Virtual Workshop on Research Writing Skills, May 2020.
- 40-hour online FDP, Data Science for All, Electronics and ICT Department, July–August 2020.
- Online course: Python for Machine Learning, Great Learning, June 2020.
- Two-week Online Short-Term Training Programme on Research Methodology, Gautam Buddha University, July 2020.

## Teaching and Academic Coordination

**Core Computer Science and Engineering Subjects**

- Programming Fundamentals and Programming in Python
- Operating Systems
- Computer Architecture and Organization
- Theory of Computation
- Data Mining and Data Warehousing
- Core Computer Science and Engineering concepts

**Specialized and Advanced Subjects**

- Artificial Intelligence
- Machine Learning and Deep Learning
- Natural Language Processing (NLP)
- Image Processing and Medical Image Analysis
- Computer Vision
- Healthcare Artificial Intelligence

**Academic Coordination and Pedagogy**

- Course coordination, curriculum and lecture planning, practical/laboratory development, lecture notes, question-bank preparation, student mentoring, and outcome-orientated teaching.

## Recent Certifications and Professional Development

- 40-hour Agentic AI Faculty Development Programme jointly associated with IIITDM Jabalpur, IIT Kanpur, MNIT Jaipur and supported by MeitY, 2026.
- Deep Learning and Computer Vision Faculty Development Programme, May 2026.
- Qualcomm AI Upskilling Certificate, 2026.
- Databricks knowledge badges in Building Single-Agent Applications, Generative AI Application Deployment and Monitoring, and Generative AI Fundamentals, 2026.
- Google Cloud Baseline Infrastructure certification.
- Faculty Development Programme on Intellectual Property Rights for Sustainable Development, Galgotias University, February 2024.
- Faculty Development Programme on Machine Learning using Python – Hands-on Approach, June 2022.
- Faculty Development Programme on Machine Learning and Data Science using Python – Hands-on Approach, Dr. B. R. Ambedkar National Institute of Technology, Jalandhar, June 2022.

## Academic Leadership and Institutional Contributions

- Faculty Coordinator for Smart India Hackathon (SIH) 2026 Internal Finale; mentored student teams and supported institutional coordination.
- Course Coordinator for Data Mining and Warehousing and academic coordination responsibilities in current teaching role.
- Faculty responsibilities involving curriculum planning, lecture planning, practical/laboratory development, question-bank preparation, mentoring, and student activities.
- Previous NAAC Criteria 5 Coordinator and SCSE Event Coordinator at Galgotias University.
- Technical Program Committee / reviewer contributions for academic conferences, including ICCSAI and IC2PCT.

## Technical Skills

**Programming and Data:** Python, SQL
**Artificial Intelligence:** Machine Learning, Deep Learning, Computer Vision, Medical Image Processing
**Deep Learning:** CNNs, EfficientNetV2, CBAM, Transformer Architectures
**Medical Imaging:** CT Image Analysis, CLAHE, Wavelet Denoising, GLCM
**Research Tools:** TensorFlow, LaTeX, Overleaf, BibLaTeX/Biber

## Research Metrics

- 25 Scopus-indexed documents
- 70 Scopus citations
- h-index: 3
- 2 patents
- 3 book chapters

## Awards and Achievements

- Ms. Farewell Award during postgraduate study.
- Best Actress Award at zonal level, 2006.
- Best Actress Award at state level, 2007.
- Awards in declamation and debate competitions at school and college level.
- Certificate of Appreciation as Reviewer for IC2PCT 2024.
- Technical Program Committee Award, ICCSAI 2023, Galgotias University.
- Felicitation by Lions Club Meerut/Bhawani in recognition of contribution as an educator.
- Poems published in the international poetry books *The Elegance of Rhyming Poetry* and *For the Love of Poetry*.

- Best Actress Award at zonal level (2006) and state level (2007).
- Awards in declamation and debate competitions at school and college level.

## Academic, Creative and Outreach Contributions

- Poems published in two international poetry books: *The Elegance of Rhyming Poetry* and *For the Love of Poetry*.
- Faculty/event coordination and organization of academic and cultural events.
- Participation in social-service activities through Parijat Youth Organization.

## Academic Profiles

- ORCID: 0000-0002-7940-0650
- Google Scholar: Aanchal Vij
- LinkedIn: Aanchal Vij
- ResearchID: Aanchal Vij

## Personal Information and Declaration

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|---|---|
| **Name** | Aanchal Vij |
| **Date of Birth** | 04 February 1992 |
| **Languages** | English, Hindi |
| **Current Location** | Greater Noida, Uttar Pradesh, India |
| **aanchal.vij@iilm.edu** | aanchal.vij@iilm.edu |
| **ORCID** | 0000-0002-7940-0650 |

**Declaration:** I hereby declare that the information provided in this curriculum vitae is true and correct to the best of my knowledge. I understand that the academic and publication details should be supported by appropriate institutional, publisher, indexing, or certification records wherever required.

**Signature:** _____ **Last Updated:** October 31, 2026

Aanchal Vij

## References

Available upon request.